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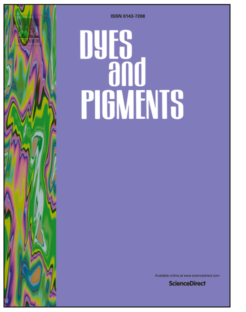

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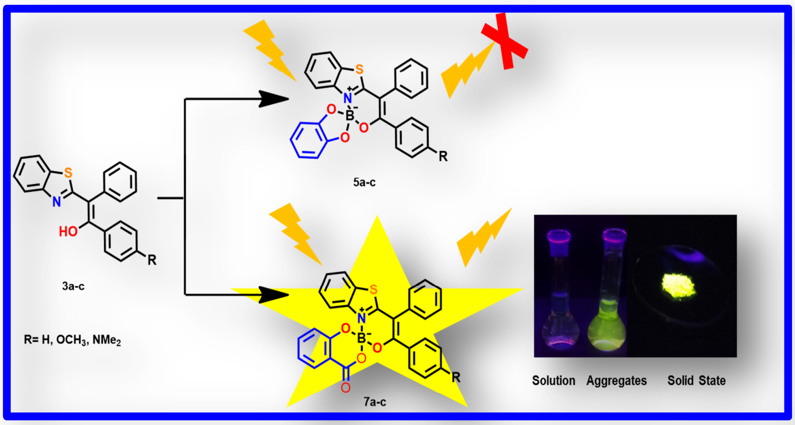

Novel, solid-state, highly fluorescent *β*-ketoiminate spiroborates with aggregation induced emission

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Abstract

Novel catechol and salicyl-spiroborates containing *β*-ketoiminate ligand showing keto-enol tautomerism were synthesized. The fluorescent properties of the synthesized spiroborates were investigated. These spiroborates showed intense fluorescence in the solid-state and weak fluorescence in solution. These spiroborates showed prominent aggregation induced emission activity because of restricted intramolecular rotations of the aryl rings. The salicyl spiroborates showed better fluorescence property than the corresponding catechol spiroborates. The structural, electronic and spectroscopic properties were evaluated by using density functional theory (DFT).

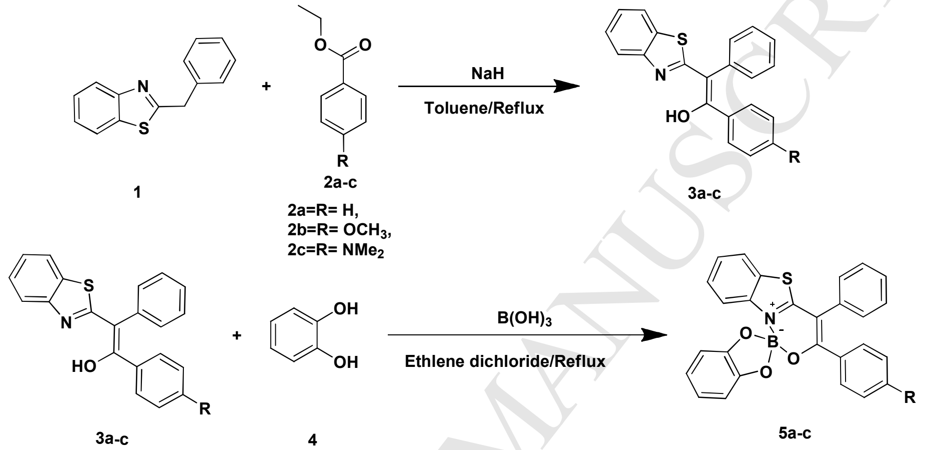
Keywords: *β*-ketoiminate, Spiroborate, Aggregation induced emission (AIE), Solid-state emission

1. Introduction

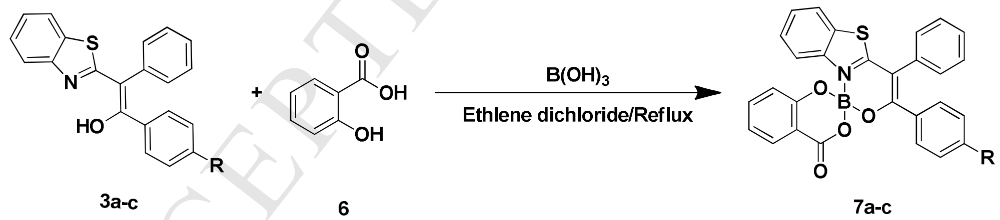
The design and synthesis of the solid-state emissive organoboron complexes have gained great attention in optoelectronic applications[1–7]. The boron complexes mostly boron dipyrromethene (BODIPY) dyes[8–11] have outstanding fluorescent properties. The BODIPY dyes rarely show fluorescence in the solid-state[12] due to the their planarity and intermolecular π–π-stacking interactions, which result in concentration quenching in the solid-state itself. This phenomenon is known as aggregation caused quenching (ACQ)[13–18]. Only few boron complexes are known for exhibiting photoluminescence in their solid state and aggregation induced emission (AIE)[19] phenomenon which includes systems like pyridomethene[20], *β*-diketone[21], *β*-iminoketone[22], *β*-ketoiminate[19] and *β*-enaminone[23]. These organoboron complexes including BODIPY are extensively used in chemosensing[24–30], biolabelling[31,32], dye-sensitized solar cells[33,34] and in photodynamic therapy[35,36].

Spiroborates are tetracoordinate boron complexes with two π-planes oriented orthogonal to one another[37]. These complexes are particularly unique as the electron delocalization over the spiroboron centre arises from π-donation of the oxygen atoms into the empty *p*-type orbital of the boron atom[37]. Spiroborate derivatives are used primarily as an electrolyte in lithium batteries[38–42]. They also offer different advantages as these materials are usually nontoxic, inexpensive and thermally, chemically, and electrochemically stable[43]. Spiroborates show intrastrand charge-transfer transitions[44] and electrochemical stability[45,46]. The fluorescent spiroborate e.g. boron-dipyrrin oligomers[47,48] used in metal ion recognition and *β*-diketoboronates[49], are used as laser dyes and semiconductor[50] due to their higher stability and strong π-electron accepting properties.

Recently we have reported β-ketoiminate based BF_2-complexes which showed solid-state fluorescence and AIE activity[51]. In extension of our work, we have designed and synthesized tautomeric β-ketoiminate based catechol spiroborate and salicyl spiroborate complexes. Their photophysical study showed intense solid-state emission and AIE activity. Density functional computations were used to understand the geometry and electronic properties of the synthesized spiroborates.



Scheme1. Synthesis of catechol spiroborates compounds.



Scheme2. Synthesis of salicylspiroborate compounds.

2 Experimental

2.1 Materials and methods

All reagents and other chemicals were obtained from commercial suppliers and used without further purification. Nuclear magnetic resonance spectra were recorded on Bruker 300 MHz or Varian 300M instruments with TMS as an internal standard. UV-Visible spectra were performed with a Perkin-Elmer Lamda-25 spectrophotometer at room temperature Fluorescence spectra were performed with a Varian Cary Eclipse fluorescence spectrofluorometer at room temperature. Particle size determined using Nano plus DLS particle size analyser.

2.2 Synthesis 3a-c

(Z)-2-(Benzo[d]thiazol-2-yl)-1,2-diphenylethenol (3a)

Sodium hydride (60 wt% in oil, 1.6 g, 40 mmol) was added to a toluene (100 mL) solution of 2-benzylbenzo[*d*]thiazole **1** (2.25 g, 10 mmol) and ethyl benzoate **2a** (1.5 g, 10 mmol) at room temperature. The solution was refluxed overnight. After cooling to 0 to 5°C, aq. NH_4Cl was added to the reaction mixture and extracted with ethyl acetate. The extract was dried over $MgSO_4$ and concentrated in vacuum. Column chromatography of the residue on silica gel (Hexane: ethyl acetate 80:20) gave **3a** as a pale yellow powder (1.8g, 57%): mp (118-120°C); FT-IR (cm^{-1}): 1584, 1560, 1489, 1455, 1249, 1159, 1084, 1059, 779. ^{1}H NMR (400 MHz, $CDCl_3$) δ 13.90 (s, 1H), 7.66 (d, J = 7.8 Hz, 1H), 7.41 (d, J = 8.0 Hz, 1H), 7.29 (m, 10H), 7.18 (t, J = 7.4 Hz, 2H). ^{13}C NMR (101 MHz, $CDCl_3$) δ 166.20, 165.67, 148.74, 139.46, 135.29, 134.39, 132.04, 129.51, 129.36, 129.26, 129.16, 129.08, 128.81, 128.42, 127.67, 127.37, 124.75, 124.24, 117.88, 110.60, 101.08. HRMS (m/z): Calculated for $C_{21}H_{16}NOS$ Calculated 330.0947(M+H), found 330.0944(M+H).

(Z)-2-(Benzo[d]thiazol-2-yl)-1-(4-methoxyphenyl)-2-phenylethenol (3b)

Synthesized by same procedure as **3a** using ethyl 4-methoxybenzoate **2b** and compound **1**

Yellowish green powder (Yield= 55%): mp 140-142°C; FT-IR (cm^{-1}): 1597, 1560, 1506, 1452, 1437, 1295, 1248, 1173, 1082, 1058, 1028, 835,746, 723. ^{1}H NMR (500 MHz, $CDCl_3$) δ 15.02 (s, 1H), 7.84 (d, J = 8.1 Hz, 2H), 7.69 (d, J = 7.9 Hz, 1H), 7.45 (t, J = 7.6 Hz, 1H), 7.37 (m, 4H), 7.31 (d, J = 8.6 Hz, 2H), 7.27 (t, J = 7.6 Hz, 1H), 6.71 (d, J = 8.7 Hz, 2H), 3.77 (s, 3H). ^{13}C NMR (126 MHz, $CDCl_3$) δ 193.79, 173.30, 169.71, 163.93, 163.73, 160.13, 152.41, 150.42, 138.07, 136.68, 136.10, 131.94, 131.92, 131.51, 130.71, 129.30, 129.08, 128.73, 128.57, 128.26, 128.22, 127.93, 126.43, 125.88, 124.92, 123.95, 122.98, 121.52, 121.12, 119.84, 114.01, 113.03, 106.81, 57.96, 55.52, 55.19. HRMS (m/z): Calculated for $C_{22}H_{21}NO_2S$ $(M+H)^+$ 360.1053, Observed 360.1049$(M+H)^+$.

(Z)-2-(Benzo[d]thiazol-2-yl)-1-(4-(dimethylamino)phenyl)-2-phenylethenol (3c)

Synthesized by same procedure as **3a** using **2c** and compound **1** Yield= 61%): mp 156-158°C; FT-IR (cm^{-1}): 1657, 1582, 1524, 1436, 1370, 1192, 1154, 1059, 760, 707. ^{1}H NMR (400 MHz, CDCl3) δ 8.03 – 7.95 (m, 3H), 7.81 (d, J = 8.0 Hz, 1H), 7.56 (d, J = 7.5 Hz, 2H), 7.44 – 7.36 (m, 2H), 7.35 – 7.19 (m, 3H), 6.59 (d, J = 9.0 Hz, 2H), 6.56 (s, 1H), 2.99 (s, 6H).13C NMR (101 MHz, CDCl3) δ 192.98, 170.29, 153.70, 152.39, 137.50, 136.24, 132.10, 131.43, 130.48, 129.09, 128.69, 127.85, 125.70, 124.73, 123.34, 122.89, 121.46, 110.79, 110.73, 57.41, 39.89. HRMS (m/z): Calculated for $C_{23}H_{21}N_2OS$ Calculated 373.1370 (M+H), found 373.1376 (M+H).

2.3 Synthesis of catechol-spiroborates (5a-c)

Synthesis of 3,4-diphenylspiro[benzo[4,5]thiazolo[3,2-c][1,3,2]oxazaborinine-1,2'-enzo[d][1,3,2]dioxaborol]-10-ium-12-uide (***5a)***

Compound **3a** (329mg, 1mmol) and catechol **4** (110mg, 1mmol) were dissolved in ethylene dichloride (5mL). After that boric acid (68mg, 1.1mmol) was added and heated the reaction mass for next 5 hours at reflux. After that reaction mass was cooled and solvent was evaporated. Residue purified by column chromatography (Hexane: ethyl acetate 80:20) to give yellow non fluorescent solid compound **5a** (300mg, 67%): mp 228-230°C; FT-IR (cm^{-1}): 1597, 1583, 1548, 1481, 1461, 1436, 1368, 1322, 1231, 1213, 1112, 1051, 1026, 943, 878, 745. ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.63 (dd, *J* = 6.4, 2.3 Hz, 1H), 7.48 – 7.41 (m, 5H), 7.41 – 7.31 (m, 5H), 7.28 (d, *J* = 7.3 Hz, 1H), 7.15 (t, *J* = 7.7 Hz, 2H), 6.94 (dd, *J* = 5.1, 3.2 Hz, 2H), 6.88 (dd, *J* = 5.3, 3.5 Hz, 2H). ^{13}C NMR (101 MHz, $CDCl_3$) δ 172.36, 163.17, 150.53, 143.99, 135.11, 134.01, 131.32, 130.63, 130.09, 129.69, 129.05, 128.83, 128.45, 127.72, 125.76, 121.57, 119.98, 118.28, 109.89, 106.75. HRMS (m/z): Calculated for $C_{27}H_{18}BNO_3S$ (M+H)+ 448.1178, Observed 448.1175.

Synthesis of 3-(4-methoxyphenyl)-4-phenylspiro[benzo[4,5]thiazolo[3,2-c][1,3,2]oxazaborinine-1,2'-benzo[d][1,3,2]dioxaborol]-10-ium-12-uide ***(5b)***

Synthesized by same procedure as **5a** using compound **3b** and **4** (Yield= 63%): mp.164-168°C; FT-IR (cm^{-1}): 1601, 1578, 1544, 1454, 1435, 1366, 1309, 1257, 1257, 1232, 1176, 1045, 950, 822, 739, 695. ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.61 (d, *J* = 6.7 Hz, 1H), 7.50 – 7.45 (m, 3H), 7.41 (d, *J* = 9.0 Hz, 2H), 7.39 – 7.29 (m, 4H), 7.18 (d, *J* = 7.3 Hz, 1H), 7.00 – 6.85 (m, 1H), 6.66 (d, *J* = 9.0 Hz, 2H), 3.75 (s, 3H).^{13}C NMR (101 MHz, $CDCl_3$) δ 162.85, 161.66, 150.62, 144.78, 144.05, 135.62, 132.22, 131.41, 129.84, 129.04, 128.35, 128.24, 127.03, 125.51, 125.23, 121.49, 119.91, 118.09, 113.19, 109.88, 77.33, 77.01, 76.70, 55.28. HRMS (m/z): Calculated for $C_{28}H_{20}BNO_4S$ $(M+H)^+$ 478.1284, Observed 478.1286.

Synthesis of 3-(4-(dimethylamino)phenyl)-4-phenylspiro[benzo[4,5]thiazolo[3,2-c][1,3,2]oxazaborinine-1,2'-benzo[d][1,3,2]dioxaborol]-10-ium-12-uide ***(5c)***

Synthesized by same procedure as **5a** using compound **3c** and **4** (Yield= 70%): mp.- 282-284°C; FT-IR (cm^{-1}): 1605, 1558, 1518. 1481, 1453, 1421, 1365, 1232, 1115, 1047, 940, 816, 747, 722, 636. ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.55 (d, *J* = 7.6 Hz, 1H), 7.48 (t, *J* = 6.2 Hz, 3H), 7.40 (dd, *J* = 7.5, 1.8 Hz, 2H), 7.36 (d, *J* = 9.2 Hz, 2H), 7.29 (d, *J* = 3.7 Hz, 2H), 7.23 – 7.13 (m, 1H), 6.93 (m, 2H), 6.91 – 6.85 (m, 2H), 6.38 (d, *J* = 9.2 Hz, 2H), 2.94 (s, 6H). ^{13}C

NMR (101 MHz, $CDCl_3$) δ 171.88, 163.46, 151.87, 150.80, 144.16, 136.47, 132.29, 131.57, 129.84, 128.80, 128.44, 128.07, 124.96, 121.32, 120.43, 119.71, 117.66, 110.44, 109.79, 104.65, 39.87. HRMS (m/z): Calculated for $C_{29}H_{23}BN_2O_3S$ $(M+H)^+$ 491.1600, Observed 491.1603.

2.4 Synthesis of salicylspiroborate (7a-c)

*Synthesis of 4'-oxo-3,4-diphenyl-4'H-spiro[benzo[4,5]thiazolo[3,2-c][1,3,2]oxazaborinine-1,2'-benzo[d][1,3,2]dioxaborinin]-10-ium-12-uide**(7a)***

Synthesized by same procedure as **5a** using compound **3a** and **6** (Yield= 65%): mp 180 – 184°C; FT-IR (cm^{-1}): 16981612,1580, 1560, 1477, 1446, 1320, 1303, 1258, 1242, 1133, 1110, 1050, 1020, 955, 860, 750, 684. ^{1}H NMR (400 MHz, $CDCl_3$) δ 8.16 (dd, *J* = 7.8, 1.4 Hz, 1H), 7.68 (d, *J* = 8.2 Hz, 2H), 7.57 (t, *J* = 6.9 Hz, 1H), 7.44 (d, *J* = 6.8 Hz, 4H), 7.40 (dd, *J* = 4.2, 1.2 Hz, 1H), 7.37 (d, *J* = 7.3 Hz, 4H), 7.29 (d, *J* = 7.4 Hz, 1H), 7.15 (t, *J* = 7.7 Hz, 2H), 7.10 (t, *J* = 7.6 Hz, 1H), 7.06 (d, *J* = 8.2 Hz, 1H). ^{13}C NMR (101 MHz, $CDCl_3$) δ 172.96, 163.59, 163.28, 158.18, 143.37, 135.66, 134.93, 133.87, 131.31, 130.73, 130.45, 129.95, 129.73, 129.13, 129.03, 128.42, 127.80, 125.95, 121.82, 120.65, 118.42, 118.07, 115.22, 106.78. HRMS (m/z): Calculated for $C_{28}H_{18}BNO_4S$ $(M+H)^+$ 476.1127, Observed 476.1123.

*Synthesis of 3-(4-methoxyphenyl)-4'-oxo-4-phenyl-4'H-spiro[benzo[4,5]thiazolo[3,2-c][1,3,2]oxazaborinine-1,2'-benzo[d][1,3,2]dioxaborinin]-10-ium-12-uide **(7b)***

Synthesized by same procedure as **5a** using compound **3b** and **6** (Yield= 70%): mp.186-180°C; FT-IR (cm^{-1}): 1719, 1602, 1579, 1544, 1461, 1448, 1301, 1255, 1236, 1181, 1112, 1027, 750, 692. ^{1}H NMR (400 MHz, $CDCl_3$) δ 8.16 (dd, *J* = 7.8, 1.3 Hz, 1H), 7.64 (d, *J* = 8.2 Hz, 2H), 7.56 (t, *J* = 6.9 Hz, 1H), 7.47 (d, *J* = 2.9 Hz, 3H), 7.40 (d, *J* = 7.4 Hz, 1H), 7.37 (d, *J* = 6.4 Hz, 2H), 7.33 (d, *J* = 8.9 Hz, 3H), 7.10 (t, *J* = 7.5 Hz, 1H), 7.05 (d, *J* = 8.2 Hz, 1H), 6.64 (d, *J* = 9.0 Hz, 2H), 3.73 (s, 3H). ^{13}C NMR (101 MHz, $CDCl_3$) δ 172.87, 163.40, 163.18, 161.70, 158.25, 143.38, 135.62, 135.38, 132.04, 131.34, 130.42, 129.85, 129.09, 129.03, 128.85, 128.26, 128.22, 126.01, 125.67, 125.29, 121.74, 120.56, 118.43, 117.82, 115.23, 113.26, 105.80, 55.27. HRMS (m/z): Calculated for $C_{29}H_{20}BNO_5S$ $(M+H)^+$ 506.1233, Observed 506.1221.

Synthesis of 3-(4-(dimethylamino)phenyl)-4'-oxo-4-phenyl-4'H-spiro[benzo[4,5]thiazolo[3,2-c][1,3,2]oxazaborinine-1,2'-benzo[d][1,3,2]dioxaborinin]-10-ium-12-uide ***(7c)***

Synthesized by same procedure as **5a** using compound **3c** and **6** (Yield= 65%): mp 230-234°C; FT-IR (cm^{-1}): 1712, 1605, 1557, 1451, 1418, 1369, 1258, 1199, 1035, 947, 824, 751, 691. ^{1}H NMR (400 MHz, $CDCl_3$) δ 8.16 (dd, *J* = 7.8, 1.5 Hz, 1H), 7.59 (d, *J* = 8.1 Hz, 2H), 7.55 (d, *J* = 7.2 Hz, 1H), 7.46 (m, 5H), 7.37 (t, *J* = 7.4 Hz, 2H), 7.32 (m, 3H), 7.09 (t, *J* = 7.9 1H), 7.05 (d, *J* = 8.3 Hz, 1H), 6.37 (d, *J* = 9.2 Hz, 1H), 2.93 (s, 6H). ^{13}C NMR (101 MHz, $CDCl_3$) δ 172.42, 163.77, 163.69, 158.45, 151.90, 143.50, 136.26, 135.49, 132.13, 131.53, 130.39, 129.86, 128.87, 128.58, 127.98, 125.13, 121.56, 120.36, 120.17, 118.48, 117.42, 115.32, 110.48, 104.62, 39.85. HRMS (m/z): Calculated for $C_{30}H_{23}BN_2O_4S$ $(M+H)^+$ 519.1550, Observed 519.1552.

2.5 Fluorescence quantum yields (ϕ_f)

The fluorescence quantum yields were calculated using Quinine sulphate as standard (ϕ_f =0.54 in 0.1M H_2SO_4) as the standard in various THF–water mixtures by using the steady-state comparative method[52–55]

$$\phi_s = \phi_{st} \, X \, (I_s / I_{st}) \, X \, (A_{st}/A_s) \, X \, (\eta_s^2/\eta_{st}^2)$$

ϕ_s is the fluorescence quantum yield of the sample, ϕ_{st} is the fluorescence quantum yield of the standard, A_{st} and A_s represent the absorbance of the standard and the sample at the excitation wavelength, respectively. I_{st} and I_s are the integrated emission of the standard and the sample, respectively, and η_{st} and η_s are the solvent refractive indices of the standard and the sample.

2.6 Computational methodology:

Density functional theory (DFT) was performed using B3LYP[56,57] functional and the def2-TZVP (valence triple-zeta-plus polarization)[58,59] as basis set for corresponding operations, using the Turbomole-V6.5 program[60–62]. Time dependant density functional (TDDFT) calculations were also performed with same B3LYP functional and the def2-TZVP basis set.

3. Result and discussion

3.1 Design and Synthesis

In an extension of our recently published work[51], here we have replaced the two fluorine atoms with catechol and salicyl fragments because spiroborates consist of tetra-coordinated boron with two π-planes oriented orthogonal to one another which helps in minimising π-π stacking and thereby enhances the emission in the solid-state. The unique electron delocalization over the spiro boron centre arises from π-donation of the oxygen atoms into the empty p-type orbital of the boron atom and intrastrand charge-transfer transitions[37,44]. Synthesis of the catechol and salicyl spiroborates was achieved in 2 steps. Condensation of 2-benzylbenzo[*d*]thiazole (**1**) with an ethyl benzoate derivative (**2a-c**), in presence of sodium hydride in toluene gave compounds (**3a-c**). Reaction of compounds **3a-c** with either catechol and salicylic acid in presence of boric acid using ethylene chloride as solvent gave the desired aryl spiroborates **5a-c** (Scheme 1.) and **7a-c** (Scheme 2.) respectively. All of the synthesized spiroborates were purified by column chromatography and characterized by using ^{1}H and ^{13}C NMR spectroscopy and mass spectrometry.

The synthesised compounds **3a-c** shows keto-enol tautomerization. The compound **3a** predominantly exists in the enol form and **3c** exists in its keto [19,51,63]. Whereas the ratio of keto: enol tautomer of **3b** was found to be 1:2.5 established by ^{1}H NMR in $CDCl_3$. However on complexation reaction the ^{1}H NMR signal for enolic proton (δ 13.90) of **3a** and alpha proton (δ 6.56) of **3c** (keto form) disappears indicating the formation of spiroborates. Similarly in **3b** both enolic (δ 15.02) and alpha proton (δ 6.57) disappears on spiroborate formation.

3.2 Absorption and fluorescence properties

The UV-Visible and emission spectra of the catechol and salicyl-spiroborates (**5a-c** and **7a-c)** were recorded in tetrahydrofuran. Amongst the compound **5a-c**, only **5c** fluoresces with low quantum yield[53–55] (ϕ_f =0.028) attributed to the presence of a strongly electron donating dimethylamino group (Fig.1) while compounds **5a** and **5b** are non-fluorescent. Compounds 7**a-c** fluoresces weakly in THF solution with lower quantum yields (Fig.2, Table 1) while Compound **5c** and **7c** exhibited bathochromic shift due to electron donating (dimethylamino) group. Their weak fluorescence in solution is consequence of the free rotatable aryl ring across the conjugated C-C single bonds in solution[13]. However it is well known that these types of molecules have the ability to fluoresce in their aggregated state where the free rotation of the conjugated aryl ring is restricted[54]. This restricted intramolecular rotation results in enhancement of fluorescence in their aggregates and the phenomenon is known as aggregation induced emission (AIE).

Fig.1. (a) Absorption spectra of the **5a-c, (b)** normalised absorption and emission spectra of the **5c in** THF (1 X 10^{-5}M).

Fig.2. (a) Absorption spectra of the **7a-c, (b)** normalised emission spectra of the **7a-c** in THF (1X 10^{-5}M).

To study the aggregation induced emission activity, we have recorded the fluorescence spectra in different fraction of the water:THF. It was observed that among the series **5a-c** only **5c** exhibits enhancement of fluorescence in the 90% water:THF system, whereas all **7a-c** show remarkable enhancement in the fluorescence in 90% water:THF system. The higher quantum yields were observed for compounds **5c** and **7a-c** in their aggregated states at 90% water fraction (Fig.S3). Interestingly, at water fraction >90%, the mixture emits intensely and efficiently than in pure THF solution (Fig.3 & 4). The restriction of intramolecular rotation is responsible for the AIE phenomenon, which blocks the nonradiative relaxation channel and populates radiative excitations[64].

The size of different nano-aggregates of **5c** and **7a-7c** at 90% water fraction was determined by dynamic light scattering (DLS) technique. The DLS study showed that an average diameter of all the compounds **5c** and **7a-c** were in the range of 237-278 nm at 90% water fraction (Fig. S4). This suggest that at 90% of water fraction leads to formation of the nano-aggregate suspension in the THF–water mixtures resulting into enhancement of fluorescence.

Fig.3. (a) PL spectra of **5c** solutions in THF–water mixtures **(b)** Photoluminescence of **5c** versus solvent composition of the THF–water mixture.

Fig.4. (a) PL spectra of **7c** solutions in THF–water mixtures **(b)** Photoluminescence of **7c** versus solvent composition of the THF–water mixture.

The emissive properties of the catechol and salicylic acid spiroborates **5a-c** and **7a-c** showed that, among the **5a-c** only **5c** has fluorescence in solid-state and solution, whereas **5a** and **5b** do not exhibit any fluorescence. However salicyl spiroborates (**7a-c**) has strong fluorescence in solid-state and is weakly fluorescent in solution. This may be because of the more electron accepting nature of the salicylate ligand than that of electron rich catechol ligand. However among **5a-c**, fluorescent nature of **5c** may be because of strong electron donation from dimethyl-amino substituent than the catechol ligand which enhances electron accepting tendency of boron centre.

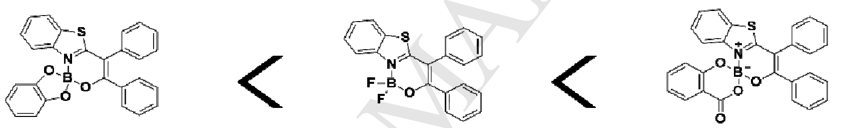
Fig.5. (a) and (b) Solid-state emissive spectra of the **5a-c** and **7a-c**

Table 1. Photophysical data of compound **5a-c** and **7a-c**

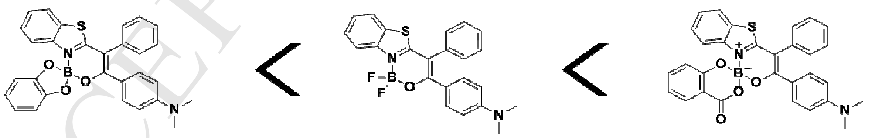
Compounds	**In Tetrahydrofuran**			ϕ^a	ϕ^b	**Solid-State**	
	λ_{Abs} (ε) (nm)	λ_{PL} (nm)	**Stoke Shift** (cm^{-1})			λ_{Abs} (nm)	λ_{PL} (nm)
5a	376 (13544)	-	-	0.000	-	-	-
5b	398 (15880)	-	-	0.000	-	-	-
5c	449 (16680)	505	4253	0.028	0.126	460	547
7a	382 (11182)	448	3857	0.011	0.09	438	490
7b	401 (13063)	481	4148	0.002	0.12	446	501
7c	464 (16937)	548	3340	0.034	0.30	461	555

[a] Quantum yield in THF
[b] Quantum yield in THF:Water (10:90) using Quinine sulphate as standard (ϕ_f=0.54 in 0.1M H_2SO_4)

On comparison of the photophysical properties with previously reported compounds **10a** and **10c**[51] with **5a, 5c, 7a** and **7c** it was observed that compounds having catechol (**5a, 5c**) and salicyl (**7a, 7c**) chelates showed a red-shift in absorption relative to **10a** and **10c**. Similarly compound **7a** and **7c** exhibited higher quantum yield in solution as well as in the aggregates whereas lower quantum yield by compound **5c**. Amongst all **10a, 10c, 5a, 5c, 7a** and **7c** compound **5a** doesn't exhibit fluorescence in solution and in its aggregated form. The photophysical data suggest that compounds with a catechol chelate (5a-5c) are non-fluorescent unlike the BF2 (**10a, 10b**) and salicyl (**7a-7c**) chelated derivatives (**Scheme 3**) with the exception of **5c**. Amongst all the compounds **7a-7c** exhibits high fluorescent quantum yield in aggregated form due to presence of electro withdrawing salicyl chelate and reduced planarity as a result of boron spiro centre.



	5a	10a	7a
λ_{Abs} (nm)	376	360	382
λ_{PL} (nm)	-	446	448
Φ_{Sol}	-	0.014	0.011
Φ_{Agg}	-	0.065	0.09



	5c	10c	7c
λ_{Abs} (nm)	449	446	464
λ_{PL} (nm)	505	507	548
Φ_{Sol}	0.028	0.01	0.034
Φ_{Agg}	0.126	0.142	0.3

Scheme 3 Molecular structure of the Organoboron compounds. (The < sign indicates the order of increase in the quantum yield)

3.3 Computational study

The ground state geometries of compound **5a-c** and **7a-c** were optimised by DFT using B3LYP functional and the def2-TZVP basis set. The optimised geometry showed twisting of ring (A) and (B) in compound **5c** by 74.01 and 27.1°, similarly in compound **7c** 73.36 and 28.49°. The twisting of ring (A) is quite more than that of ring (B) and may be due to the steric hindrance of adjacent aryl ring (B) and benzothiazolyl moiety. The spiroborates having catechol and salicylic ring (c) are orthogonal to boron centre as shown in the **Fig 6** and **Fig S5-S6.**

Fig 6. DFT optimised geometry of compound **5c** and **7c** front view, **5'c** and **7'c** top view.

To analyse the electronic spectra we have performed TDDFT calculations with B3LYP functional and the def2-TZVP basis set in tetrahydrofuran. The calculated λ_{max} values are in accordance with the experimental values**.** The calculated λ_{max}, the main orbital transition, and oscillator strength (*f*) are shown in **Table S1** and theoretical results showed similar trend in case of electronic properties. The HOMO/LUMO energies of the **5a-5c** were quite higher than that of **7a-7c**, suggesting that the boron catechol complex (**5a-5c**) have less electron accepting property. In the case of **5a-5c**, absorptions are mostly due to the transition from HOMO-1 to LUMO. The HOMO orbitals of **5a-5b** delocalized over catechol ring and LUMO orbitals are delocalized over the thiazole boron chelated part of molecules (**Fig.7**.). Whereas HOMO of **5c** localised over the *N,N*-dimethyl part and LUMO is localised on thiazole boron chelated part which leads to effective charge transfer attributing to the fluorescence and a large stoke shift. While in case of **7a-7c** the absorption are due to HOMO to LUMO transition. The electron density at HOMO of **7a** and **7b** is distributed over the whole molecule and in **7c** it is distributed over *N,N*-dimethyl part. In LUMO, electron density is distributed over thiazole-Boron chelated part (**Fig.8**). The predicted dipole moment for **5c** and **7c** were larger, which showed charge separation between the dimethyl-amino group and Boron chelate (**Table S1**) with lowering in the band gap[65].

Fig. 7 Frontier molecular orbital of compound **5a-c**

Fig. 8 Frontier molecular orbital of compound **7a-c**

4. Conclusion

In conclusion, herin we have reported novel catechol and salicyl spiroborates containing benzothiazolyl-1,2-diaryl β-ketoiminate ligand. The synthesized spiroborates are weakly fluorescent in solution with low fluorescence quantum yield but showed intense fluorescence in their aggregated states. All the salicyl-spiroborates exhibit noticeable aggregation induced emission. The salicyl spiroborates showed better emissive properties than their corresponding catechol-spiroborates.

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Tables

Table 1. Photophysical data of compound **5a-c** and **7a-c**

Compounds	**In Tetrahydrofuran**					**Solid State**	
	λ_{Abs} (ε) (nm)	**λ_{PL} (nm)**	**Stoke Shift (cm^{-1})**	ϕ^a	ϕ^b	**λ_{Abs} (nm)**	**λ_{PL} (nm)**
5a	376 (13544)	-	-	0.000	-	-	-
5b	398 (15880)	-	-	0.000	-	-	-
5c	449 (16680)	505	4253	0.028	0.126	460	547
7a	382 (11182)	448	3857	0.011	0.09	438	490
7b	401 (13063)	481	4148	0.002	0.12	446	501
7c	464 (16937)	548	3340	0.034	0.30	461	555

[a] Quantum yield in THF
[b] Quantum yield in THF:Water (10:90) using Quinine sulphate as standard (ϕ_f=0.54 in 0.1M H_2SO_4)

Figures and Captions

Scheme1. Synthesis of catechol spiroborates compounds.

Scheme2. Synthesis of salicylspiroborates compounds.

Fig.1. (a) Absorption spectra of the **5a-c, (b)** normalised absorption and emission spectra of the **5c in** THF

Fig.2. (a) Absorption spectra of the **7a-c, (b)** normalised emission spectra of the **7a-c** in THF

Fig.3. (a) PL spectra of **5c** solutions in THF–water mixtures **(b)** Photoluminescence of **5c** versus solvent composition of the THF–water mixture.

Fig.4. (a) PL spectra of **7c** solutions in THF–water mixtures **(b)** Photoluminescence of **7c** versus solvent composition of the THF–water mixture.

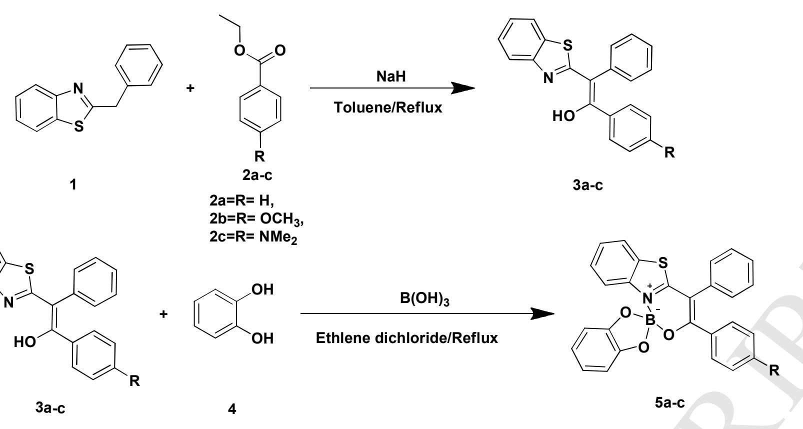
Fig.5. (a) and (b) Solid state emissive spectra of the **5a-c** and **7a-c**

Scheme 3 Molecular structure of the Organoboron compounds. (The $<$ sign indicates the order of increase in the quantum yield)

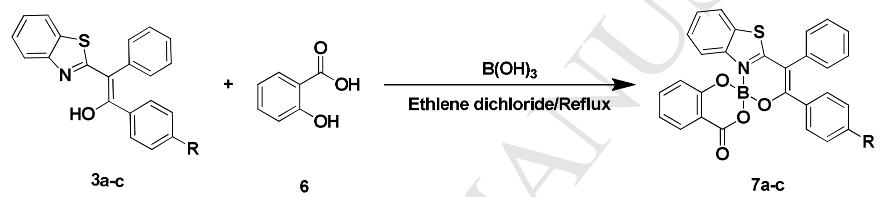
Fig 6. DFT optimised geometry of compound **5c** and **7c** front view, **5'c** and **7'c** top view.

Fig. 7 HOMOL- LUMO molecular orbital of compound **5a-c**

Fig. 8 HOMO- LUMO molecular orbital of compound **7a-c**



Scheme1. Synthesis of catechol spiroborates compounds.



Scheme2. Synthesis of salicyl spiroborates compounds.

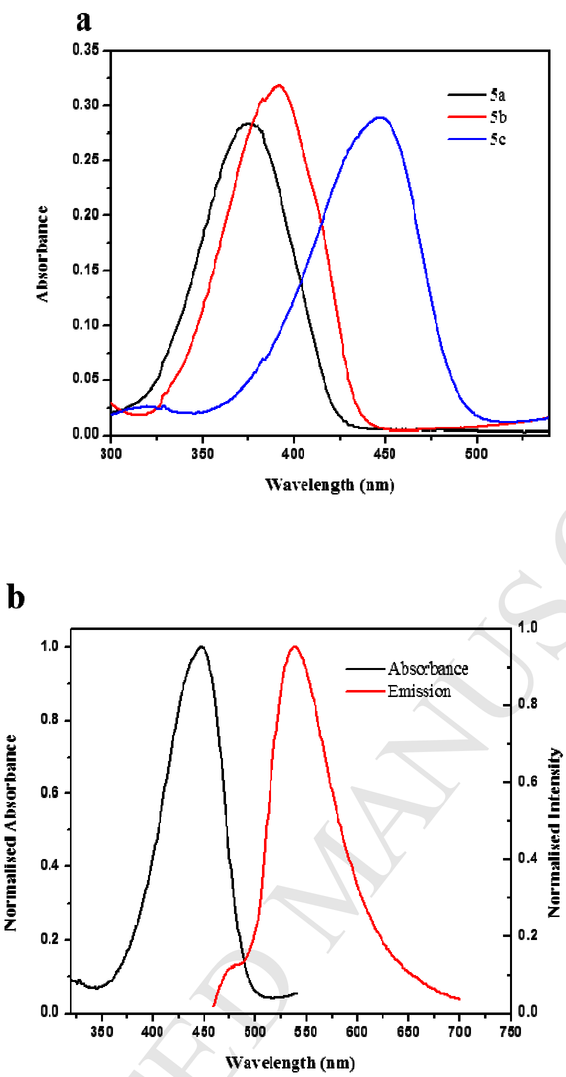


Fig.1. (a) Absorption spectra of the **5a-c, (b)** normalised absorption and emission spectra of the **5c in** THF

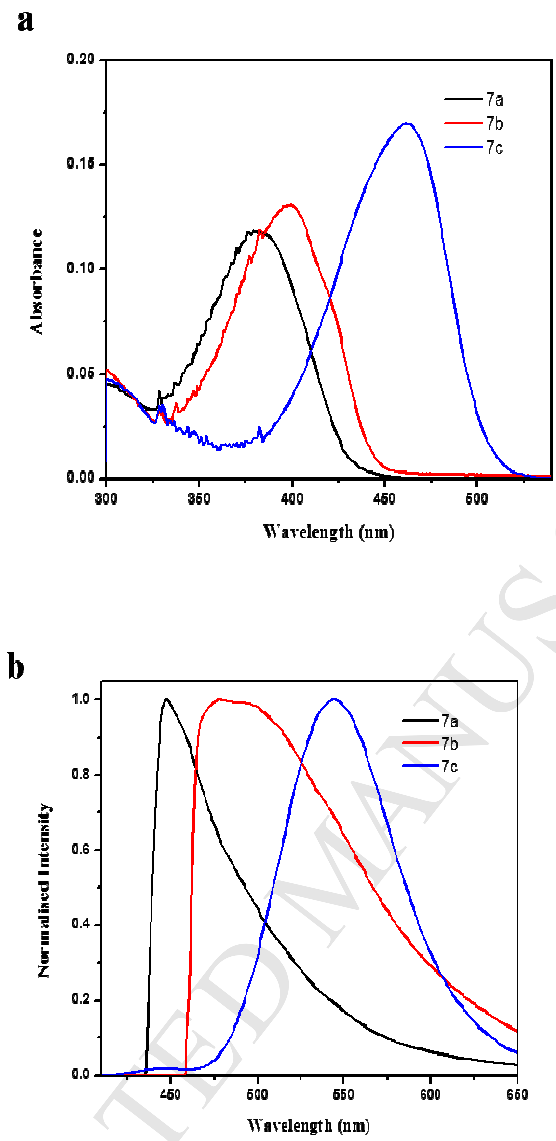
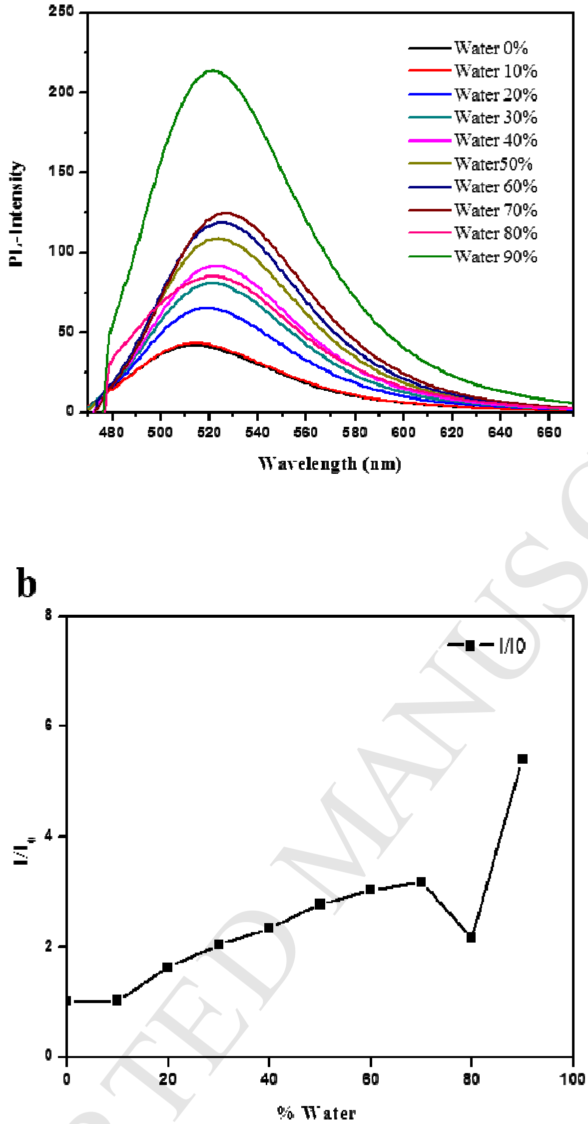


Fig.2. (a) Absorption spectra of the **7a-c, (b)** normalised emission spectra of the **7a-c** in THF

a



b

Fig.3. (a) PL spectra of **5c** solutions in THF–water mixtures **(b)** Photoluminescence of **5c** versus solvent composition of the THF–water mixture.

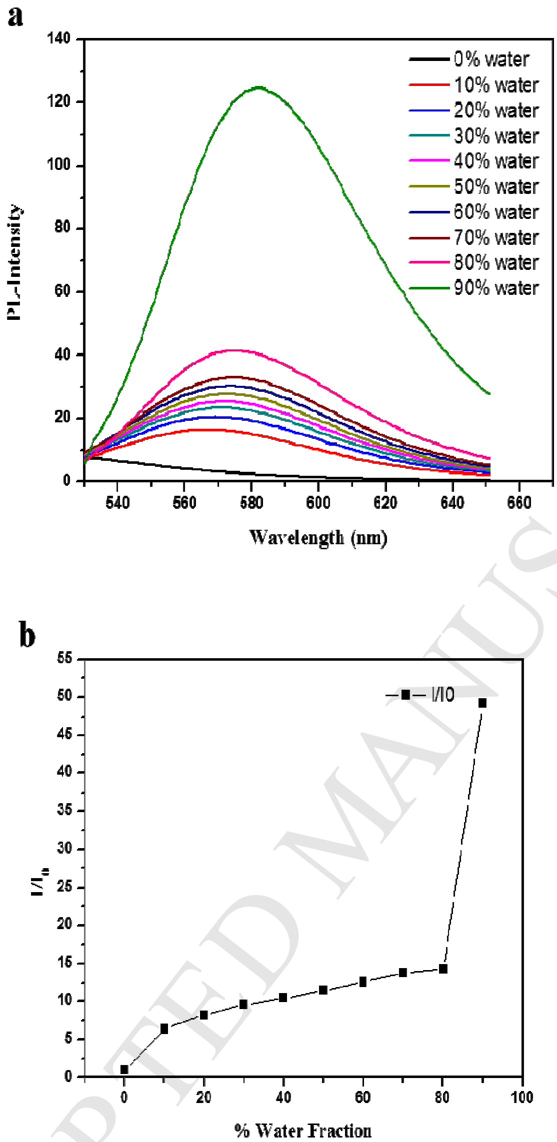
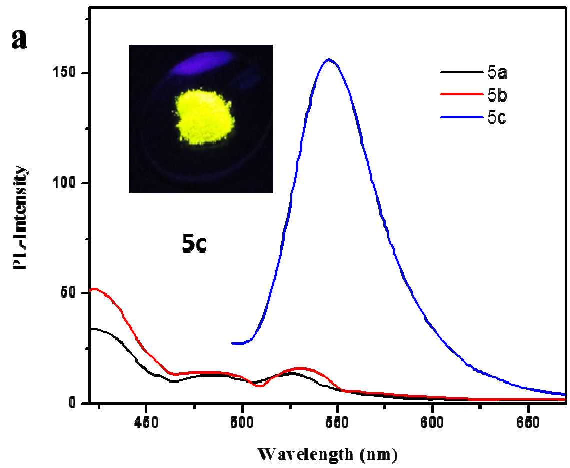


Fig.4. (a) PL spectra of **7c** solutions in THF–water mixtures **(b)** Photoluminescence of **7c** versus solvent composition of the THF–water mixture.



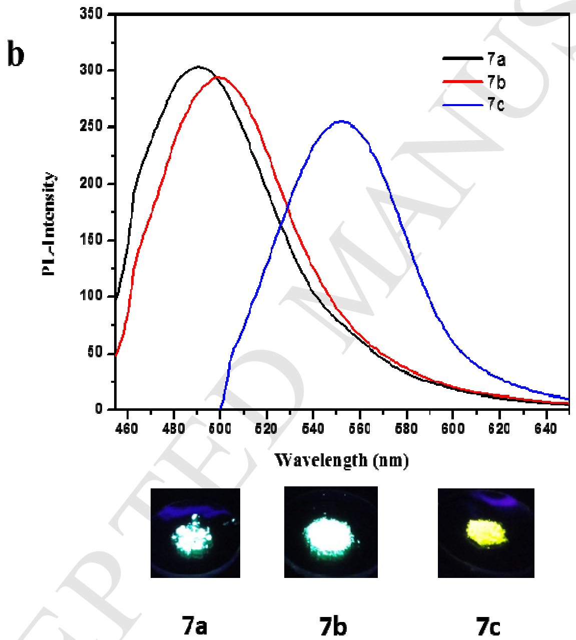
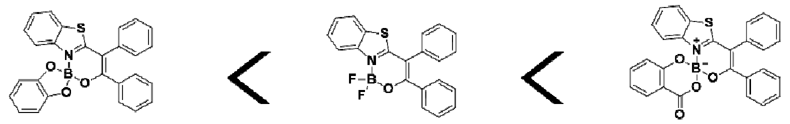
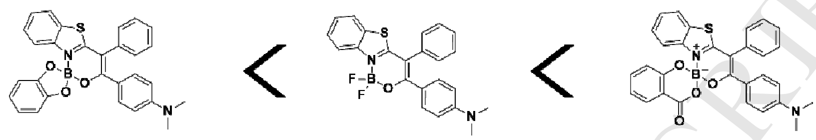


Fig.5. (a) and (b) Solid state emissive spectra of the **5a-c** and **7a-c**



	5a	10a	7a
λ_{Abs} (nm)	376	360	382
λ_{PL} (nm)	-	446	448
Φ_{Sol}	-	0.014	0.011
Φ_{Agg}	-	0.065	0.09



	5c	10c	7c
λ_{Abs} (nm)	449	446	464
λ_{PL} (nm)	505	507	548
Φ_{Sol}	0.028	0.01	0.034
Φ_{Agg}	0.126	0.142	0.3

Scheme 3 Molecular structure of the Organoboron compounds (The < sign indicates the order of increase in the quantum yield)

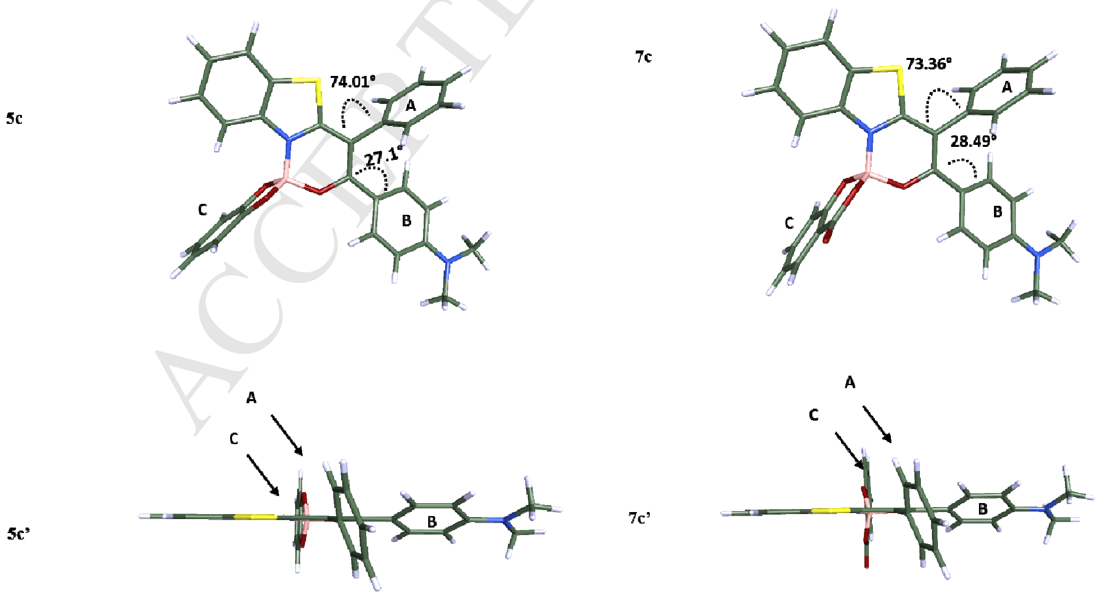


Fig 6. DFT optimised geometry of compound **5c** and **7c** front view, **5'c** and **7'c** top view.

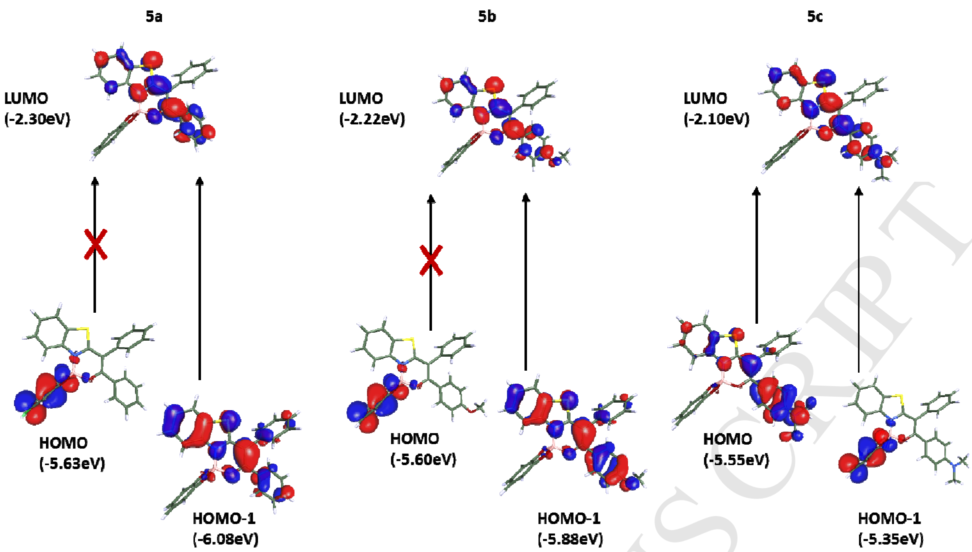


Fig. 7 HOMOL- LUMO molecular orbital of compound **5a-c**

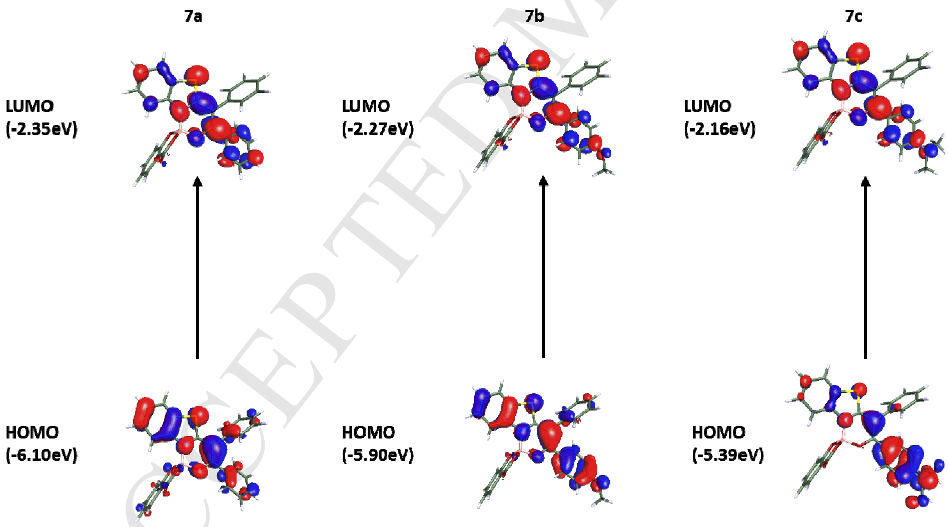


Fig. 8 HOMO- LUMO molecular orbital of compound **7a-c**

Highlights

1. Novel benzothiazolyl-1,2-diaryl β-ketoiminate spiroborates were synthesized

2. The synthesized spiroborates shows intense fluorescence in the solid-state

3. These compounds exhibits aggregation induced emission activity